Vertex, Axis, Intercepts, and Opening Direction
Answer Key

Algebra 1

Quick Answers

1. (a) axis of symmetry = 3, (b) vertex y-value = −4
2. (a) x-intercepts = −2, 4, (b) y-intercept = −8
3. (b) axis of symmetry = 2, —
4. (a) axis of symmetry = −3, (b) vertex y-value = −5
5. (a) x-intercepts = −6, 2, (b) y-intercept = −12
6. (a) vertex y-value = −9, (b) y-intercept = 7
7. (a) x-intercepts = 2, 4, (b) y-intercept = −8
8. (a) axis of symmetry = 2, (b) vertex y-value = −1, (c) x-intercepts = 1, 3, —
9. (a) axis of symmetry = 3, (b) vertex y-value = −7, —
10. (b) axis of symmetry = 2, (c) vertex y-value = 18, (d) x-intercepts = −1, 5, —

What is verified

6 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 3, 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1
HSA-SSE.B.3 – *problems 2, 5, 6, 7*
HSF-IF.C – *problems 1, 3, 4, 8, 9, 10*
Difficulty 1–5 of 5 across 25 tagged checks.

Common wrong answers

1. *If they got* -3*: used b over 2a without the leading minus sign*
4. *If they got* -12*: stopped at the numerator and never divided by twice the leading coefficient*

1. $f(x) = x^2 - 6x + 5$. (a) Axis of symmetry. (b) Vertex y-value.
Solution (a): Here $a = 1$ and $b = -6$, so $x = -\frac{b}{2a} = -\frac{-6}{2}$. $\boxed{x = 3}$
Solution (b): The vertex sits on that line, so evaluate there: $3^2 - 6(3) + 5 = 9 - 18 + 5$. The vertex is $(3, -4)$. $\boxed{y = -4}$

2. $g(x) = x^2 - 2x - 8$. (a) x-intercepts. (b) y-intercept.

Solution (a): Set the output to zero and factor: $x^2 - 2x - 8 = (x+2)(x-4)$, since $(+2)(-4) = -8$ and $2 + (-4) = -2$. A product is zero only when a factor is.

$$\boxed{x = -2 \quad \text{or} \quad x = 4}$$

Solution (b): The y-intercept is the output at input 0: $0 - 0 - 8$ — the constant term. $\boxed{y = -8}$

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3. $h(x) = -3x^2 + 12x - 5$. (a) Opening direction, justified. (b) Axis of symmetry.

(a) — instructor-judged. A correct answer says the parabola opens downward and gives the reason: the coefficient of the squared term is -3, which is negative, so large inputs in either direction drive the output far below the vertex. Full credit for the direction plus the sign of the leading coefficient as the reason; partial credit for the direction alone.

Solution (b): $a = -3$ and $b = 12$, so $x = -\frac{12}{2(-3)} = -\frac{12}{-6}$. $\boxed{x = 2}$

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4. $f(x) = 2x^2 + 12x + 13$. (a) Axis of symmetry. (b) Vertex y-value.

Solution (a): $a = 2$ and $b = 12$, so $x = -\frac{12}{2(2)} = -\frac{12}{4}$. $\boxed{x = -3}$

Common wrong answer: -12 comes from stopping at the numerator and never dividing by $2a$.

Solution (b): $2(-3)^2 + 12(-3) + 13 = 18 - 36 + 13$. The vertex is $(-3, -5)$. $\boxed{y = -5}$

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5. $f(x) = x^2 + 4x - 12$. (a) x-intercepts. (b) y-intercept.

Solution (a): $x^2 + 4x - 12 = (x+6)(x-2)$, since $(+6)(-2) = -12$ and $6 + (-2) = 4$. Setting each factor to zero *flips* its sign.

$$\boxed{x = -6 \quad \text{or} \quad x = 2}$$

Common wrong answer: 6 and -2 come from reading the numbers straight off the factors without changing their signs.

Solution (b): The output at input 0 is the constant term. $\boxed{y = -12}$

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6. $f(x) = (x-4)^2 - 9$. (a) Vertex y-value. (b) y-intercept.

Solution (a): A square is never negative, so the smallest the expression can be is when $(x-4)^2 = 0$, at $x = 4$. There the value is $0 - 9$, and the vertex is $(4, -9)$. $\boxed{y = -9}$

Solution (b): At input 0: $(0-4)^2 - 9 = 16 - 9$. $\boxed{y = 7}$

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7. $f(x) = -x^2 + 6x - 8$. (a) x-intercepts. (b) y-intercept.

Solution (a): Factor out -1 first: $-(x^2 - 6x + 8) = -(x-2)(x-4)$. The overall -1 can never be zero, so the zeros come from the two brackets.

$$\boxed{x = 2 \quad \text{or} \quad x = 4}$$

Solution (b): At input 0: $-0 + 0 - 8$. $\boxed{y = -8}$

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8. $f(x) = x^2 - 4x + 3$. (a) Axis. (b) Vertex y-value. (c) x-intercepts. (d) Labelled sketch.

Solution (a): $a = 1$, $b = -4$, so $x = -\frac{-4}{2}$. $\boxed{x = 2}$

Solution (b): $2^2 - 4(2) + 3 = 4 - 8 + 3$. The vertex is $(2, -1)$. $\boxed{y = -1}$

Solution (c): $x^2 - 4x + 3 = (x - 1)(x - 3)$.

$$\boxed{x = 1 \quad \text{or} \quad x = 3}$$

(d) — instructor-judged. The sketch must show an upward-opening parabola whose lowest point sits at $(2, -1)$, whose curve crosses the horizontal axis at $x = 1$ and $x = 3$, and whose vertical axis of symmetry $x = 2$ is drawn and labelled through the vertex. Full credit when all four features are drawn and labelled and the curve is symmetric about that line; partial credit for a correct curve with unlabelled features.

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9. $f(x) = 3x^2 - 18x + 20$. (a) Axis. (b) Vertex y-value. (c) Maximum or minimum, justified.

Solution (a): $a = 3$, $b = -18$, so $x = -\frac{-18}{2(3)} = \frac{18}{6}$. $\boxed{x = 3}$

Solution (b): $3(3)^2 - 18(3) + 20 = 27 - 54 + 20$. The vertex is $(3, -7)$. $\boxed{y = -7}$

(c) — instructor-judged. A correct answer says the vertex is a minimum, because the coefficient of the squared term is positive, so the parabola opens upward and the vertex is its lowest point; every other input gives a larger output. Full credit for the label plus the sign-of-the-leading-coefficient reason; partial credit for the label alone.

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10. $f(x) = -2x^2 + 8x + 10$. (a) Direction and vertex type. (b) Axis. (c) Vertex y-value. (d) x-intercepts.

(a) — instructor-judged. A correct answer says the parabola opens downward because the coefficient of the squared term is -2, which is negative, and therefore its vertex is a maximum rather than a minimum. Full credit for both the direction and the consequence for the vertex, each tied to the sign of that coefficient; partial credit for the direction alone.

Solution (b): $a = -2$, $b = 8$, so $x = -\frac{8}{2(-2)} = -\frac{8}{-4}$. $\boxed{x = 2}$

Solution (c): $-2(2)^2 + 8(2) + 10 = -8 + 16 + 10$. The vertex is $(2, 18)$, the highest point of the curve. $\boxed{y = 18}$

Solution (d): Factor out -2: $-2(x^2 - 4x - 5) = -2(x - 5)(x + 1)$.

$$\boxed{x = -1 \quad \text{or} \quad x = 5}$$